# Cross-Modality Transfer Learning for Knee Meniscus Segmentation via Registration-Constrained Unpaired Image Translation

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## 1. DESCRIPTION OF PURPOSE

The meniscus is a critical fibrocartilaginous structure responsible for load distribution, shock absorption, and joint stability in the knee. MRI is the gold-standard non-invasive imaging modality for meniscus tear diagnosis, enabling visualization of internal derangement, tear morphology, and associated cartilage pathology without ionizing radiation. Accurate automated meniscus segmentation in MRI supports quantitative grading of cartilage injury, longitudinal monitoring of osteoarthritis progression, and pre-surgical planning. Proton density (PD)-weighted MRI is the dominant clinical protocol for knee evaluation, offering high soft-tissue contrast and short acquisition times. However, training supervised segmentation models for PD MRI requires expert voxel-level annotation a process demanding several hours of manual delineation per volume by trained radiologists making large labeled PD datasets impractical to acquire at scale.

Dual-echo steady-state (DESS) MRI provides complementary T2 relaxometry and has been distributed alongside dense expert segmentation labels in publicly available datasets such as SKM-TEA [1]. The tissue contrast between the two sequences differs substantially: fluid and cartilage appear dark on DESS but bright on PD, with reversed contrast at tissue boundaries, making direct transfer of a DESS-trained model to PD images unsuccessful. Prior unpaired image-to-image translation methods such as CycleGAN [2] have achieved modality-level appearance transfer but produce uncontrolled spatial deformations, distorting meniscus anatomy and rendering associated segmentation masks unreliable for downstream training.

This work addresses this limitation by introducing a registration-constrained generative pipeline (RegGAN) that enforces minimum geometric deformation during DESS-to-PD translation. We evaluate whether synthetic PD images generated under anatomy-preserving constraints are sufficient to transfer meniscus segmentation knowledge from the DESS domain to real PD MRI, and whether augmenting synthetic training data with a small cohort of manually annotated real PD cases further improves segmentation on unseen clinical data.

## 2. METHOD

**Datasets Description.** DESS MRI volumes were sourced from the SKM-TEA dataset [1] (69 subjects, 512×512×160 voxels, 0.31×0.31×0.80 mm spacing), which provides dense expert segmentation labels for six tissue classes including lateral and medial meniscus. PD-weighted MRI volumes were drawn from an institutional cohort (IU dataset; 69 subjects, 384×384×36 voxels, 0.39×0.39×3.60 mm spacing). Of the 69 IU PD subjects, 32 received expert manual meniscus annotations: 15 were used for supervised fine-tuning in the synthetic and real PD configuration, and 17 were reserved as a held-out evaluation cohort, with no overlap between the two groups.

**Domain adaptation.** RegGAN extends the CycleGAN framework [2] with a lightweight differentiable registration network R, implemented as a 2D convolutional U-Net inspired by VoxelMorph [3]. The generator GAB translates DESS slices to synthetic PD; R then takes the concatenation of the synthetic PD and a real PD image as input and predicts a dense 2D displacement field. Three auxiliary loss terms constrain the predicted field: a registration similarity loss ($\lambda$=5.0) enforcing pixel-level alignment between the warped synthetic PD and real PD, a smoothness penalty ($\lambda$=10.0) on spatial gradients of the displacement field, and a magnitude penalty ($\lambda$=5.0) directly penalizing displacement amplitude. Generators and discriminators follow a ResNet-based [4] architecture with 9 residual blocks and PatchGAN [5] discriminators trained using the least-squares GAN objective [6].

**Segmentation and fine-tuning.** A publicly available 2.5D U-Net with ResNet34 encoder pre-trained on DESS for multi-class knee segmentation [7] serves as the source-domain baseline. The original 5-class head is replaced by a 3-class head (background, lateral meniscus, medial meniscus) for cross-modality transfer. In the first fine-tuning configuration (synthetic PD), only synthetic PD slices paired with DESS meniscus labels are used. In the second configuration (synthetic and real PD), 15 manually annotated real PD patients are added to training using a merged binary loss that combines lateral and medial softmax probabilities against binary ground truth. Both configurations use weighted cross-entropy (background weight 0.1, meniscus weight 1.5) combined with soft Dice loss, cosine learning rate scheduling ($1\times10^{-5}$ to $1\times10^{-8}$), and augmentation including horizontal/vertical flipping, brightness jitter (±20%), and Gaussian noise ($\sigma$=0.02), trained for 50 epochs with early stopping (patience 10).

**Evaluation.** All segmentation models were evaluated on a held-out cohort of 17 real PD patients with expert manual meniscus annotations, distinct from the fine-tuning cohort. Performance was quantified using the binary Dice coefficient (lateral and medial meniscus combined).

## 3. RESULTS AND EVALUATION

**Domain adaptation.** RegGAN achieved an FID of 164.4 between synthetic PD and real PD images, compared to 260.4 for raw DESS images evaluated against the same real PD distribution, a 37% reduction in distributional distance. The SSIM between DESS and synthetic PD was 0.357, reflecting the expected contrast shift rather than structural failure. Deformation analysis confirmed anatomy preservation throughout: the mean Jacobian determinant of the registration field was 1.000056; the rate of Jacobian folding ($\det(J) < 0$, indicating topology violations) was 0.0% across all test volumes; and the mean displacement at the meniscus region was 0.055 pixels. The KID between synthetic and real PD was 0.020 ± 0.000, further confirming distributional alignment between the translated and real PD images.

**Transfer learning and Evaluation.** Segmentation performance on the 17-patient held-out cohort is summarized in Table 1. The pre-trained DESS model applied without fine-tuning achieves a mean binary meniscus Dice of 0.00, confirming complete failure of zero-shot cross-modality transfer. Fine-tuning on synthetic PD alone improves mean Dice to 0.69, demonstrating that RegGAN-generated images provide sufficient domain alignment for meaningful label transfer. The combined approach achieves the highest performance with a mean Dice of 0.78, a relative gain of 13% over the synthetic-only baseline. Per-patient Dice scores ranged from 0.69 to 0.84 across all 17 cases. All 17 patients showed improvement over the DESS baseline with both fine-tuning configurations.

## 4. SUPPORTING IMAGES, TABLES, FIGURES

Table 1. Binary meniscus Dice score on 17 held-out real PD patients.

| Model | Training Data | Mean Dice |
|---|---|---|
| Pre-trained DESS baseline | DESS only (no PD) | 0.00 |
| Synthetic PD fine-tuning | Fake PD (RegGAN output) | 0.69 |
| Synthetic and Real PD fine-tuning | Fake PD and annotated real PD | 0.78 |


Figure 1. Overview of the proposed cross-modality domain adaptation pipeline. RegGAN translates DESS MRI (SKM-TEA, with lateral meniscus, medial meniscus, and cartilage labels) to synthetic PD MRI under minimum-deformation constraints. A pre-trained 2.5D U-Net is fine-tuned on synthetic PD and evaluated on a held-out real PD cohort.

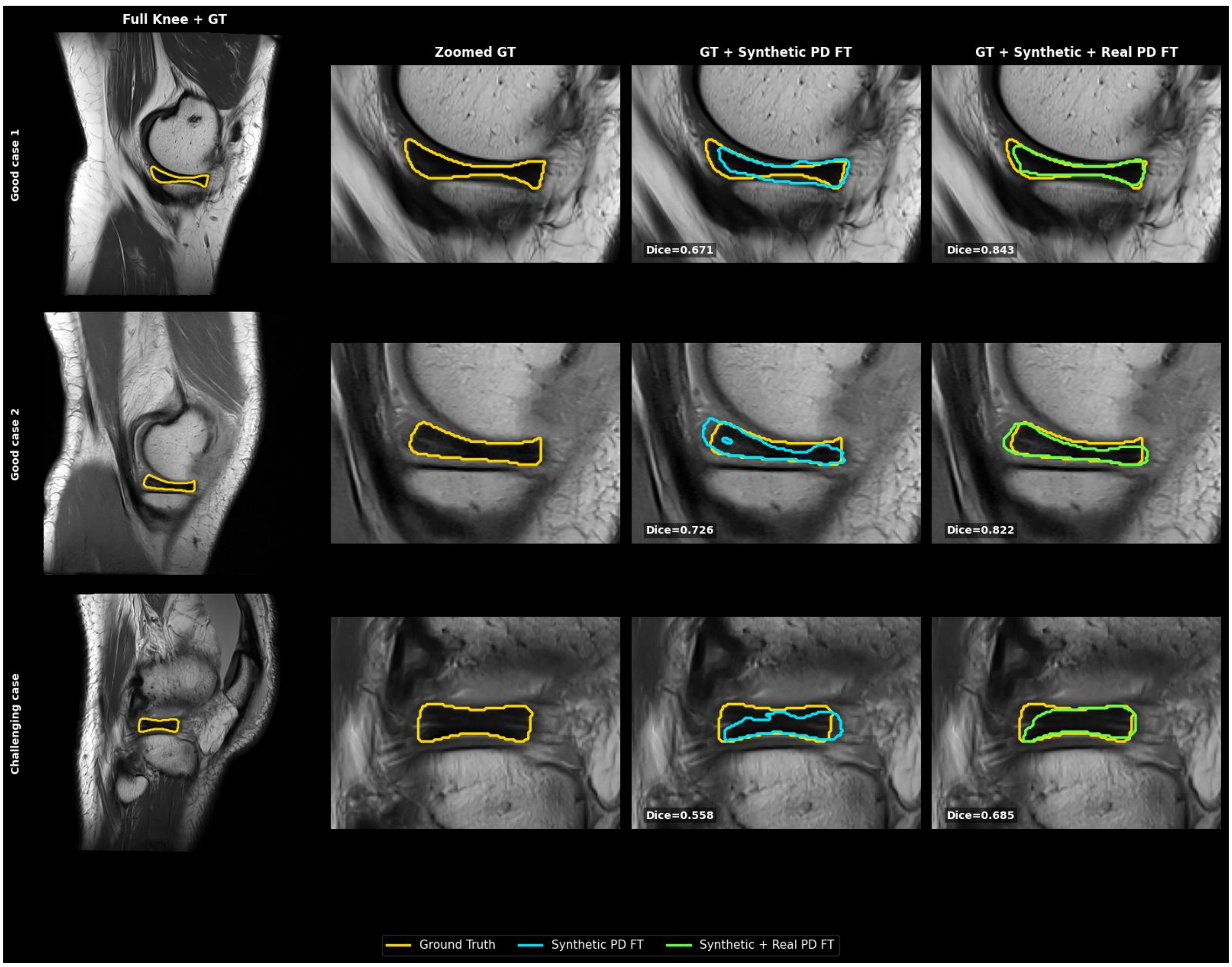


Figure 2. Meniscus boundaries on held-out real PD images. Rows: good (1–2) and challenging (3) cases; columns: full knee with ground truth (yellow), zoomed ground truth, GT + synthetic PD (cyan), GT + synthetic and real PD (green). Combined approach yields closest boundary alignment.

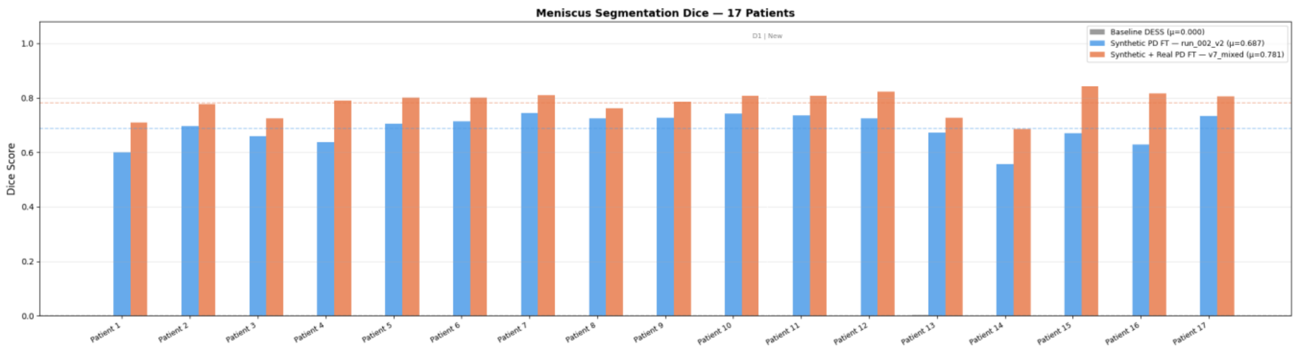


Figure 3. Binary meniscus Dice scores on 17 held-out real PD patients. Pre-trained DESS baseline (gray, μ=0.00) fails zero-shot transfer; synthetic PD fine-tuning (blue, μ=0.69) recovers segmentation; synthetic and real PD fine-tuning (orange, μ=0.78) achieves highest Dice across all patients. Dashed lines indicate per-model means.

## 5. CONCLUSIONS

This study demonstrates that registration-constrained generative domain adaptation enables effective cross-modality transfer of meniscus segmentation from DESS to PD-weighted knee MRI. The proposed RegGAN pipeline achieves near-rigid anatomical preservation during translation (mean displacement 0.055 px, 0% topology violations), allowing DESS segmentation labels to be directly reused for PD model training without paired acquisitions. On a held-out cohort of 17 expert-annotated real PD patients, fine-tuning on synthetic PD alone improves meniscus Dice from 0.00 to 0.69; augmenting with 15 annotated real PD patients further improves Dice to 0.78. These results establish anatomy-preserving GAN-based domain adaptation as a practical, label-efficient pathway for cross-modality MRI segmentation.

## 6. BREAKTHROUGH WORK

This work presents a novel anatomy-preserving cross-modality adaptation framework combining a deformation-penalized registration network within an unpaired GAN training loop with a progressive transfer learning strategy for knee meniscus segmentation. A key practical contribution is the substantial reduction in manual annotation burden: by leveraging existing publicly available DESS segmentation labels rather than requiring expert voxel-level annotation of PD MRI which demands several hours of radiologist time per volume the proposed pipeline enables effective PD segmentation with only 15 manually annotated real PD cases. By demonstrating that synthetic images generated under minimum-deformation constraints preserve the geometric fidelity needed to transfer dense segmentation labels across MRI contrasts, this study provides a reproducible and label-efficient approach applicable to other organ systems and acquisition protocol pairs where labeled source-domain data exist but target-domain annotations are scarce. **This work has not been presented or submitted for publication or presentation elsewhere.**

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